

1.Scenario Analysis:

A cloud platform uses an access control system to protect sensitive information and prevent unauthorized access. The system uses Role-Based Access Control (RBAC) to assign permissions according to users' responsibilities. Administrators have full access to manage users, configure security settings and monitor activities. Employees can access only the files and applications required for their jobs, while guests have limited access to specific shared resources.

All users must log in using strong passwords and multi-factor authentication (MFA) to improve security. The system automatically records login attempts and user activities to identify suspicious behaviour. When an employee leaves the company, their account is immediately disabled to prevent unauthorized access. Regular security reviews are conducted to ensure that users have appropriate permissions. This access control setup protects confidential information, reduces security risks and ensures that employees can perform their duties safely and efficiently.

2.concept research:

Attribute-Based Access Control (ABAC) is a security method that controls access to resources based on attributes rather than only using predefined user roles. These attributes can describe the user, resource, action, or environment. For example, an employee may access a company document only if they are in the Finance department, have the required security clearance, and are accessing it during working hours. ABAC uses policies to evaluate these attributes and make access decisions. It provides flexible and detailed control over resources, especially in cloud and large organizational environments. ABAC can reduce unauthorized access by applying specific conditions to every request. It is useful for protecting sensitive information while allowing legitimate users to access resources required for their responsibilities.

3.tool practice:

SELinux (Security-Enhanced Linux) was configured in a VirtualBox Linux environment to control access to system resources. During the practice, SELinux was enabled and its security mode was checked using system commands. Access policies were applied to restrict unauthorized users and applications from accessing protected files and services. The configuration demonstrated how SELinux provides an additional security layer beyond normal Linux permissions. Testing different access attempts showed that SELinux can block actions that violate defined security policies. This practice helped me understand how mandatory access control works in a real operating system. SELinux is useful for protecting servers and sensitive systems because it limits potential damage

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4.application practice:

An access control policy defines who can access an organization's systems, applications, and information. The policy requires every user to have a unique account and strong password. Multi-factor authentication is required for important systems. Users are assigned permissions based on their job responsibilities, following the principle of least privilege. Administrators can manage accounts and security settings, while ordinary employees can access only the resources needed for their work. Access logs are monitored to detect suspicious activities. User permissions are reviewed regularly and removed immediately when an employee leaves the organization. The purpose of this policy is to prevent unauthorized access, protect sensitive information, reduce security risks, and ensure that resources are used safely and responsibly.



ACCESS CONTROL POLICY

Secure Information System

Right people.
Right access.
Right time.

Purpose

To protect systems, data and resources by ensuring that only authorized users can access them.

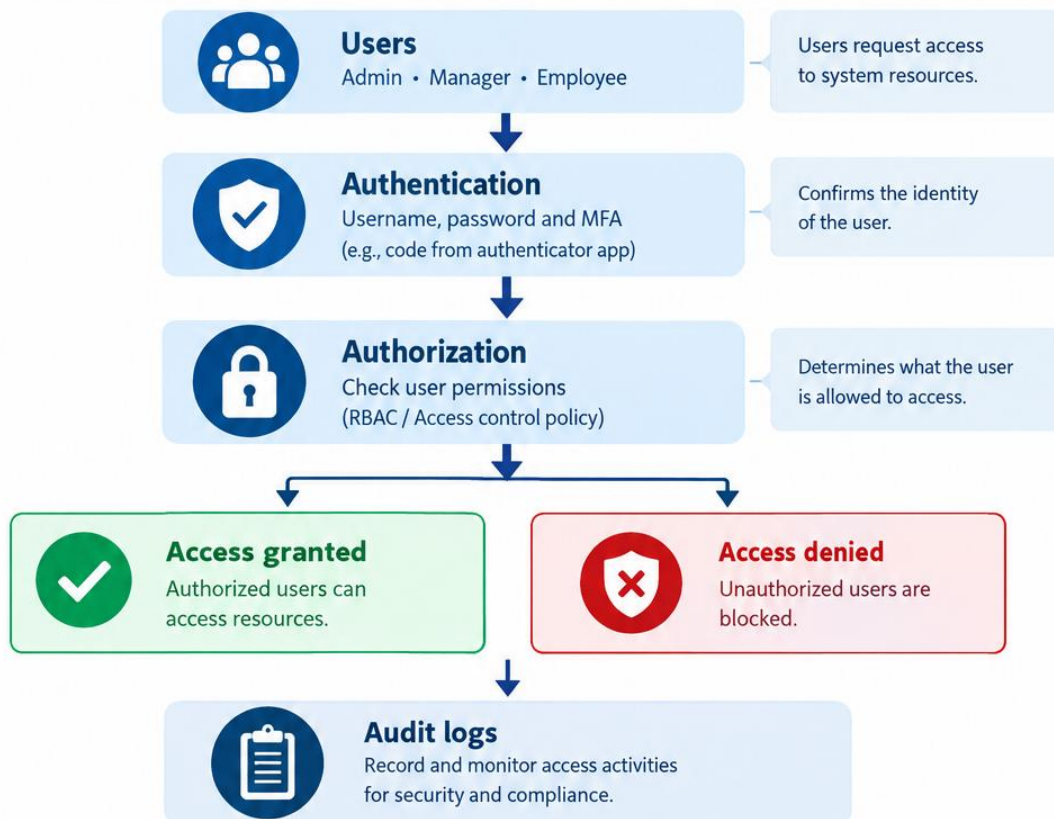
Scope

Applies to all users, devices, applications and data within the organization's information system.

Key Principles

- Least privilege
- Role-based access control (RBAC)
- Authentication and authorization
- Accountability and audit

Access Control Structure



Access Permissions

User	Permissions
 Admin	Full system access
 Manager	View and manage department records
 Employee	Access assigned files only



Policy: Users receive only the permissions necessary to perform their duties.